

ECSE 2500

Lec 11

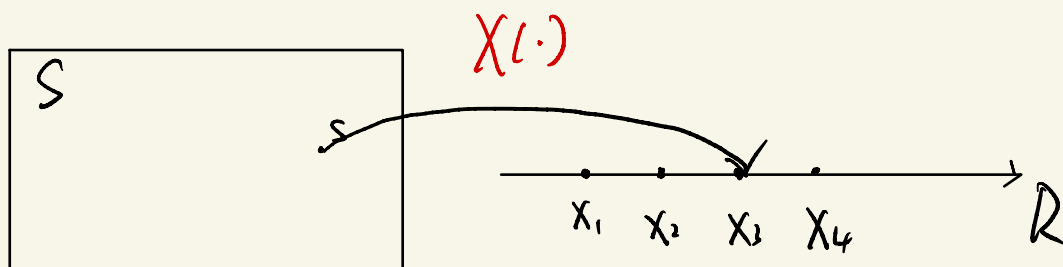
Feb 27

Topic :

Distribution
Cumulative Function (CDF)

PMF (Discrete RV) $\xrightarrow{\downarrow}$ PDF (Continuous)

□ Recall PMF for Discrete RV



All possible outcomes of X are collected into

$$S_X := \{x_1, x_2, x_3, x_4, \dots\}$$

which can be ∞ many, e.g., Geometric RV

We can write the probability mass function (PMF) as

$$P_X(k) = \begin{cases} \text{---} & , k=x_1 \\ \text{---} & , k=x_2 \\ \vdots & \end{cases}$$

$\sum_{k \in S_X} P_X(k) = 1$

□ Today we define Cumulative Distribution Function (CDF) of a RV X :

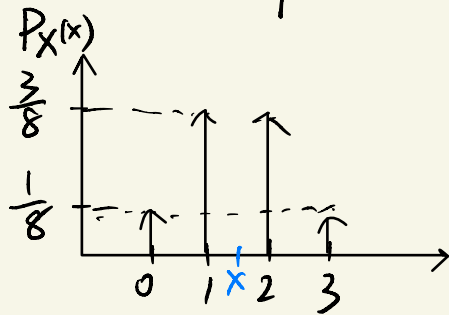
$$F_X(x) = P(X \leq x), \quad -\infty < x < \infty$$

↑ Small case
Big case Value/Outcome of
RV itself in random variable

That is, what is Probability X is less than a given value? Define CDF for any RV X (whatever discrete or continuous).

Example: Flip coin 3 times, count # of heads.

$$S_X = \{0, 1, 2, 3\}$$



Q: What will be the CDF of X like?

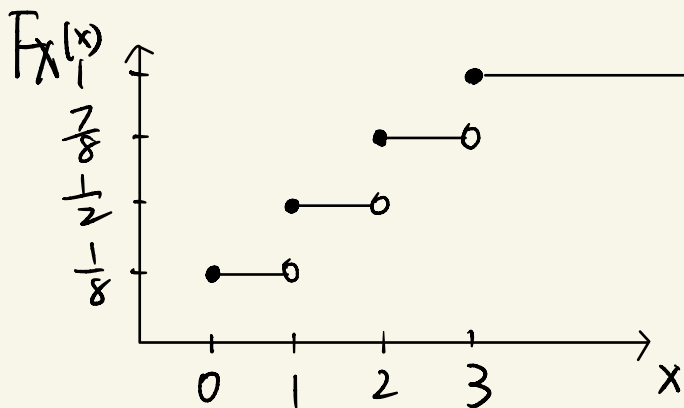
Then we can compute CDF as

$$F_X(x) = 0, \text{ if } x < 0$$

$$F_X(x) = 1, \text{ if } x > 3$$

$$F_X(1.5) = \frac{1}{8} + \frac{3}{8} = \frac{1}{2}$$

We can see in fact CDF $F_X(x)$ is a step function



We can also write

$$F_X(x) = \int_{-\infty}^{x-} P_X(y) dy,$$

we sum up the impulse function up to the x (But not including x).

Since $\int_{-\infty}^x \delta(y) dy = u(x) = \begin{cases} 1, & \forall x \geq 0 \\ 0, & \forall x < 0 \end{cases}$

we can write the equivalent form of PMF

$$P_X(x) = \sum_{x_k \in S_X} P_X(x_k) \delta(x - x_k).$$

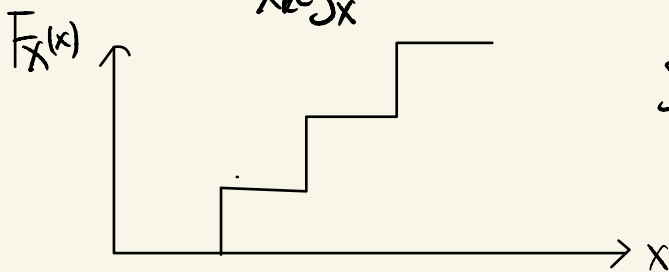
so we can write the equivalent form of CDF

$$F_X(x) = \int_{-\infty}^x P_X(y) dy$$

$$= \int_{-\infty}^x \sum_{x_k \in S_X} P_X(x_k) \delta(y - x_k) dy$$

$$= \sum_{x_k \in S_X} P_X(x_k) \int_{-\infty}^x \delta(y - x_k) dy$$

$$= \sum_{x_k \in S_X} P_X(x_k) u(x - x_k)$$



Sum of step functions

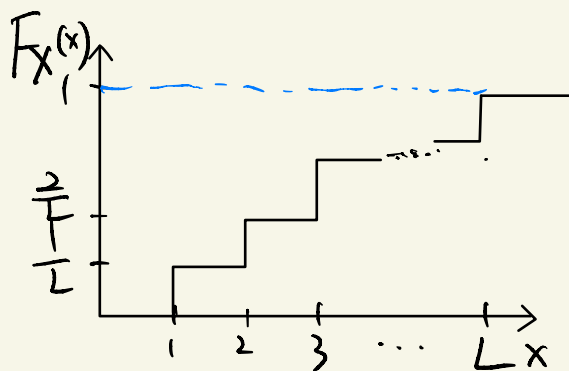
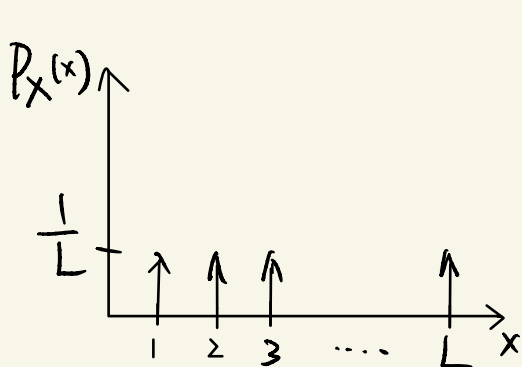
For the coin flipping example we have mentioned,
we can write the CDF as

$$F_X(x) = \frac{1}{8}u(x) + \frac{3}{8}u(x-1) + \frac{3}{8}u(x-2) + \frac{1}{8}u(x-3)$$

We have more examples on CDF for other RVs.

Example

Uniform random variable on $\{1, 2, \dots, L\}$



Example

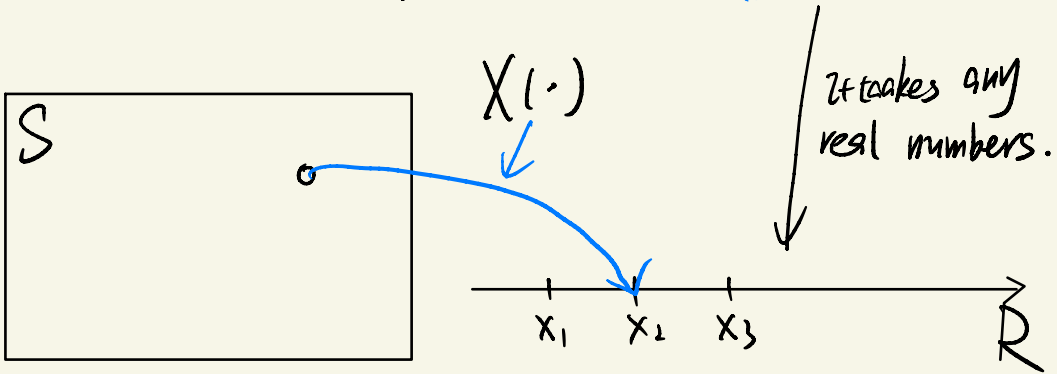
Geometric Random Variable

$$P_X(k) = (1-p)^{k-1} p, \quad k=1, 2, 3, \dots$$

$$F_X(k) = \begin{cases} 0, & \text{if } k < 1 \\ p, & \text{if } 1 \leq k < 2 \\ p + (1-p)p, & \text{if } 2 \leq k < 3 \end{cases}$$

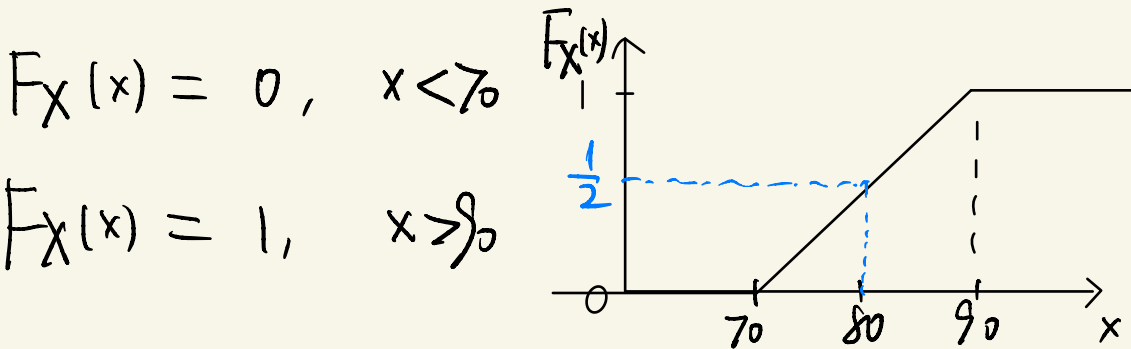
$$\begin{cases} \sum_{z=0}^{n-1} (1-p)^z p & \text{if } n \leq k < n+1 \\ = p \cdot \frac{1 - (1-p)^n}{1 - (1-p)} = 1 - (1-p)^n \end{cases}$$

□ Now we are ready to talk about continuous RVs.



Example Length of time a class lasts.

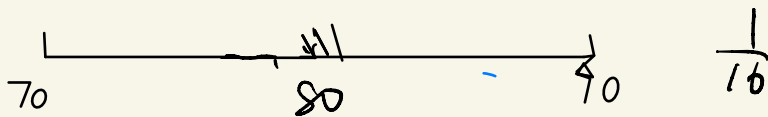
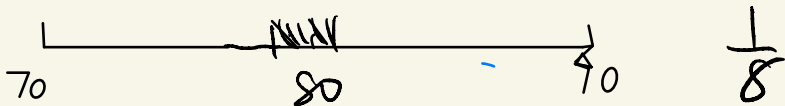
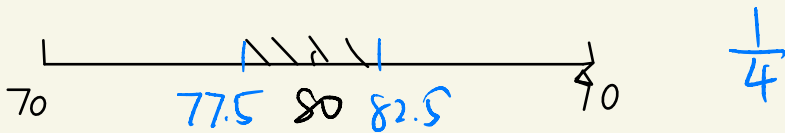
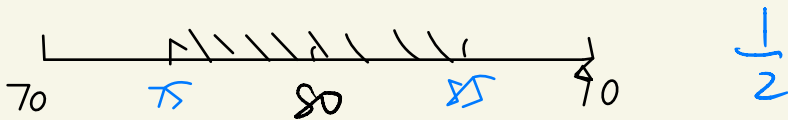
Say this is a continuous random variable uniformly distributed between 70 mins to 90 mins.



Something counterintuitive!

$$P(X = 80 \text{ min}) = 0 \quad ?$$

Think of it as follows



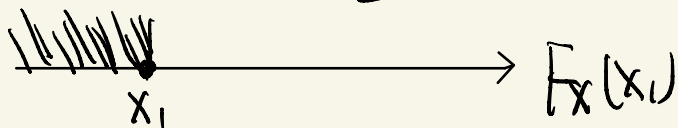
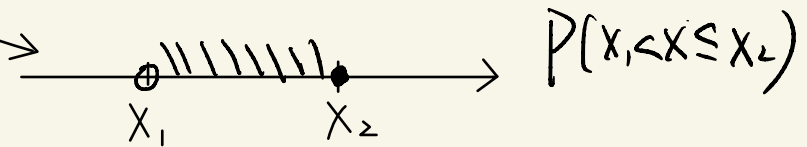
$P(X \in \text{within 1 millisecond of 80 mins})$ extremely small

$$P(X = \text{exactly 80 mins}) = 0$$

So it make sense that

$$P(x_1 < X \leq x_2) = P(X \leq x_2) - P(X \leq x_1) \\ = F_X(x_2) - F_X(x_1)$$

We can think of we can "head off" the probability of any given interval by substituting the CDF at any fixed points.



$$\lim_{\delta \rightarrow 0} P(x < X \leq x + \delta) = 0$$

Remark: Therefore, using PMF to describe continuous RV does not provide too much info.

□ Probability Density Function

So in class duration case, we can compute that

$$F_X(x) = \begin{cases} 0, & x < 70 \\ \frac{x-70}{20}, & 70 \leq x \leq 90 \\ 1, & x > 90 \end{cases}$$

Thus, we have

$$\begin{aligned} P(75 < X \leq 82) &= F_X(82) - F_X(75) \\ &= \frac{12}{20} - \frac{5}{20} \\ &= \frac{7}{20} \quad \text{Make sense!} \end{aligned}$$

Let say we want to $(a, b \in [70, 90])$

$$P(a < X \leq b) = F_X(b) - F_X(a)$$

$$\begin{array}{c} \uparrow \\ \text{Continuous RV} \end{array} = \frac{b-70}{20} - \frac{a-70}{20} = \frac{1}{20}(b-a)$$

It is also possible to describe a mixed (both discrete and continuous) RV.

Example $X = \#$ of minutes you wait for a taxi

$$P(\text{Taxi is there}) = p$$

$$P(\text{Taxi is not there}) = 1 - p$$

If taxi is there, $X = 0$

If taxi is not there, assume X is uniformly distributed in $[0, 1]$ hour.

What is the CDF of this mixed RV?

$$F_X(0) = P(X=0) = p$$

$$F_X(x) = P(X \leq x) = p + (1-p)x, \quad 0 < x \leq 1$$

$$F_X(x) = P(X \leq x) = 1, \quad x > 1$$

